

RI-RAG ARCHITECTURE & SYSTEM DESIGN

RELATIONAL-INDEXED RETRIEVAL-AUGMENTED GENERATION TECHNICAL SPECIFICATIONS

1. CORE ARCHITECTURAL CONCEPT

RI-RAG is a hybrid Retrieval-Augmented Generation architecture that separates semantic indexing from full document storage.

- **Vector Database as Semantic Entry Point:** The vector database serves strictly as the semantic entry point for queries. It stores lightweight information required for semantic retrieval, including an embedding, semantic summary, and primary/foreign keys that point to the authoritative record in a relational database.
- **Relational Database as Source of Truth:** The relational database (such as MySQL) remains the single source of truth where full document content, file metadata, semantic summaries, and detailed structured fields are securely stored, governed, and managed.
- **Memory Efficiency:** Instead of storing bulky text payloads directly in the vector database, RI-RAG uses a semantic summary of the relational content to generate an embedding. The resulting vector is stored in the vector database and used to identify the relevant relational record via relational linking.

2. VECTOR DATABASE SAMPLE

The vector database contains the lightweight semantic representation and the relationship needed to retrieve the authoritative record.

```
{ "id": "<primary_key>", "metadata": "<file_name>, <primary_key>", "documents": "  
<document semantic summary>", "embedding": "<embedding>" }
```

Note: The vector database does not store the full document content.

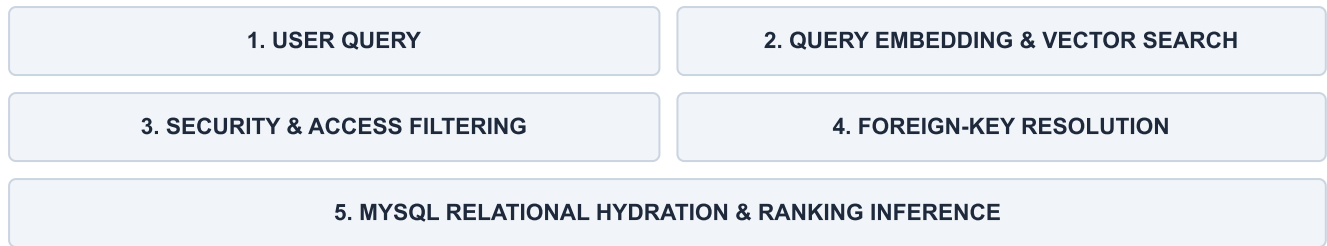
3. MYSQL TABLE SCHEMA

MySQL serves as the authoritative source of truth for the complete document payload and structured metadata.

```
CREATE TABLE policies ( file_id VARCHAR(50) PRIMARY KEY NOT NULL, file_name  
VARCHAR(255) NOT NULL, content TEXT NOT NULL, semantic_summary TEXT NOT NULL, country  
VARCHAR(255) );
```

The full content remains available in the relational layer and can be retrieved through the relationship established between the vector record and the corresponding relational record.

4. SYSTEM ARCHITECTURE & WORKFLOW



Key Workflow Components

- **Semantic Entry Point:** The vector database handles initial query ingestion and embedding similarity scanning.
- **Relational Hydration:** Upon scanning embedding potential, the system follows the key relationship to detect and disclose the right content from the relational database.
- **Entry-Only Storage:** The vector database is used strictly for entry queries and index lookups, not for output storage.
- **Ingestion & Sync (Sniffer):** New records and updates are tracked and synchronized via a custom database sniffer pipeline.
- **Hydration & Ranking Engine:** Python execution logic coordinates directly with the relational database through dynamic SQL execution scripts to fetch payloads.
- **Access Control & Filtering:** Security metadata fields are appended during vector database ingestion to enforce fine-grained access policies.

SPECIAL THANKS & BIBLIOGRAPHY

Gratefully inspired by concepts explored in *"Vector Databases Demystified"* by Abhinav Verma (Published on Medium).